

Practical – 7: Process Address Space and Memory Segments

Aim

To write a C program that prints the addresses of code, global, static, heap, and stack variables and analyze the Linux process address space using /proc/PID/maps.

```
#include <stdio.h>
#include <stdlib.h>

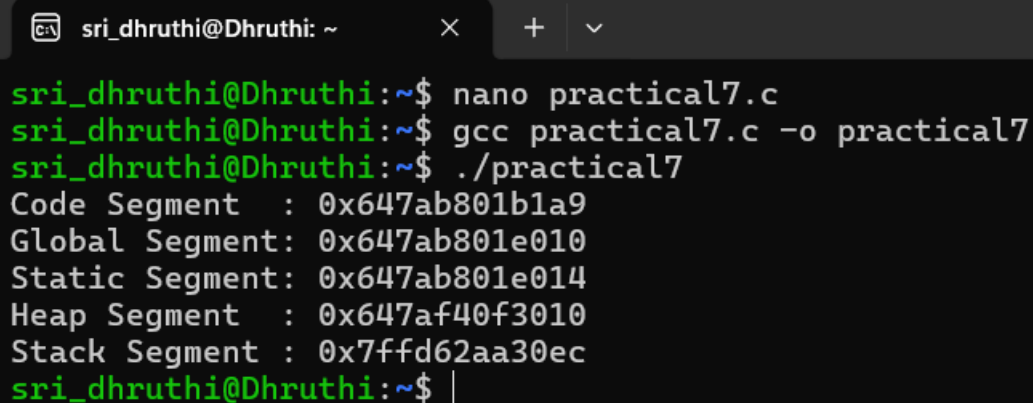
int global = 100;
static int s = 50;

int main()
{
    int stack = 10;
    int *heap = (int *)malloc(sizeof(int));

    printf("Code Segment   : %p\n", (void *)main);
    printf("Global Segment: %p\n", (void *)&global);
    printf("Static Segment: %p\n", (void *)&s);
    printf("Heap Segment   : %p\n", (void *)heap);
    printf("Stack Segment  : %p\n", (void *)&stack);

    free(heap);

    return 0;
}
```



The terminal window shows the user 'sri_dhruthi@Dhruthi' running the following commands: 'nano practical7.c', 'gcc practical7.c -o practical7', and './practical7'. The output of the program is displayed, showing the memory addresses for the Code, Global, Static, Heap, and Stack segments. The addresses are: Code Segment: 0x647ab801b1a9, Global Segment: 0x647ab801e010, Static Segment: 0x647ab801e014, Heap Segment: 0x647af40f3010, and Stack Segment: 0x7ffd62aa30ec.

```
sri_dhruthi@Dhruthi: ~$ nano practical7.c
sri_dhruthi@Dhruthi: ~$ gcc practical7.c -o practical7
sri_dhruthi@Dhruthi: ~$ ./practical7
Code Segment   : 0x647ab801b1a9
Global Segment: 0x647ab801e010
Static Segment: 0x647ab801e014
Heap Segment   : 0x647af40f3010
Stack Segment  : 0x7ffd62aa30ec
sri_dhruthi@Dhruthi: ~$ |
```